# Unsupervised Language Independent Genetic Algorithm Approach to Trivial Dialogue Phrase Generation and Evaluation

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**Abstract.** This paper describes an ongoing work on the generation and the automatic evaluation of trivial dialogue phrases. The described system uses a genetic algorithm (GA)-like transfer approach and n-grams frequency information obtained from the Web in order to automatically generate and evaluate phrases. The experiments present promising results, showing to achieve 78.51% of the *user understandability* of the generated and evaluated *good phrases* for English, and 79.06% for Spanish, and indicating the portability of the approach.

**Keywords:** GA-like transfer approach, trivial phrases, Web frequency information, language independent.

## 1 Introduction

The problematic domain of human-computer conversation (HCC) has been of particular interest to NLP-researchers, prompting them to develop conversational systems that range from applications to goal specific computer spoken dialogue, e.g., Jupiter, providing a telephone-based conversational interface for international weather [1], airline travel information systems [2], restaurant guides [3], telephone interfaces to emails or calendars [4], and so forth, to attempts to simulate human trivial dialogue - chat - e.g. ELIZA [5].

In this regard natural language generation as a sub-field of computational linguistics and as a core of HCC has withdrawn a good deal of research. From the *canned text* approach to the *template filling* approach the applications and limitations of language generation have been widely studied. One of the most famous examples of conversational systems, ELIZA [5], uses the template filling approach to generate the system's response to a user input, however it tends to become repetitive and monotonous, falling to sustain a coherent chat for long.

In recent research Inui et al. [6] have used a corpus based approach to language generation for dialogue system. Due to its flexibility and applicability to open domain, the corpus based approach might be considered more robust than the template filling approach when applied to dialogue systems. Other HCC systems, e.g. ALICE by Wallace [7], Jabberwacky by Carpenter [8] apply as well a corpus

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based approach to natural language generation in order to retrieve system's trivial dialogue responses to user inputs.

However, the creation of the hand crafted knowledge base, that is to say, a dialogue corpus, is a highly time consuming and hard to accomplish task[1]. In this paper we describe a work in progress for the unsupervised automatic generation and evaluation of a trivial dialogue phrases database. A trivial dialogue phrase is defined as an expression used by a chatbot program as the answer of a user input. A genetic algorithm (GA)-like transfer method is used to generating the trivial dialogue phrases for the creation of a natural language generation (NLG) knowledge base. The automatic evaluation of a generated phrase is performed by producing n-grams and retrieving their frequencies from the World Wide Web (WWW). The results obtained after applying the algorithm to Spanish and English are shown.

## 2 System Overview and Related Work

We apply a GA-like transfer approach to automatically generate new trivial dialogue phrases from given *seed phrases*. GA has been applied to information retrieval (IR) [9] -generating syntactic grammar rules and tag, dialogue system [10] - finding an optimal grammar which places natural language sentences input into a group of outputs clusters, and so forth, see [11] for a survey on GA/evolutionary computing together with other NLP techniques applications.

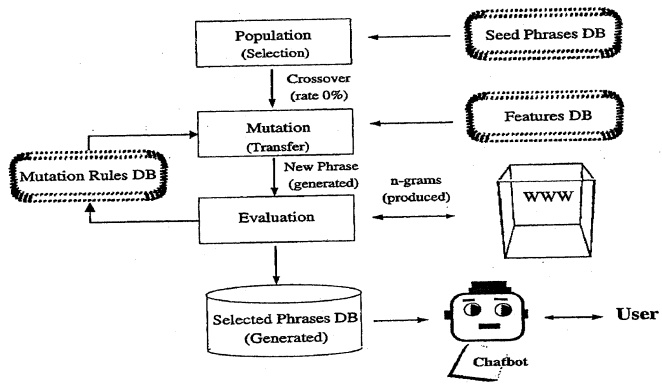



**Fig. 1.** System Overview

Blasband's application of GA to the automatic generation of grammar for a spoken dialogue system [10] was limited since the concepts expressed in every utterance along with their value needed to be determined manually. A transfer

[1] The creation of the ALICE chatbot database (ALICE brain) has cost more that 30 researchers, over 10 years work to accomplish. The Jabberwacky database is being developed since 1988 (on the Web since 1997) http://www.alicebot.org/superbot.html http://alicebot.org/articles/wallace/dont.html http://feeling.jabberwacky.com:8081/j2about

approach to language generation has been used by Arendse [12], where a sentence is being *re-generated* through word substitution. Problems of erroneous grammar or ambiguity are solved by referring to a lexicon and a grammar, re-generating substitutes expressions of the original sentence, and the user deciding which one of the generated expressions is correct. Our method differs in the application of a GA-like transfer process in order to automatically insert new features on the selected seed phrase, and in the application of the WWW in order to automatically evaluate the newly generated phrase, attempting to reduce the load of handwork. We assume the automatically generated database of trivial phrases is desirable as a knowledge base for open domain dialogue systems. The system general overview is shown in Fig. 1 and details of each process are given in Sec.3.

## 3 GA-Like Transfer Approach

### 3.1 Initial Population Selection

In the process of selection of the initial population a small number of phrases is chosen randomly from the Seed Phrases DB[2]. This is a small database created beforehand that contains phrases used during real human-human trivial dialogues. These are the seed phrases used during the regeneration.

For the experiment the English Seed Phrases DB contained phrases extracted from real human-human trivial dialogues obtained from the corpus of the University of South California [13] and from the hand crafted ALICE [7] database. The Spanish Seed Phrases DB contained dialogue phrases manually obtained from Spanish chat rooms. The initial population is then formed by a number of seed phrases randomly selected between one and the total number of seeds in the database. No evaluation is performed to this initial population.

### 3.2 Crossover

In order to ensure a language independent method our algorithm does not use syntactic information (part of speech tagging). Hence, as to avoid the distortion of the seed phrase due to completely blind crossover, in our system the crossover rate was selected to be 0%. It is found in the literature examples of GA applications to NLP were the mutation parameter has been solely the basis, e.g. [9]. In our approach the generation of the new phrase is as well given solely by the mutation process explained below.

### 3.3 Mutation

During the mutation process, each one of the seed phrases selected in the initial population is mutated at a rate of $1/N$, where $N$ is the total number of words in the phrase (for Spanish we consider a word to be an *article+noun* combination when it appears). The mutation is performed through a transfer process, using

[2] In this paper DB stands for database.

the Features DB. For the experiment this database contained 100 different features for Spanish and 110 features for English (the word "features" refers here to nouns, adjectives, adverbs and verbs).

The word to be replaced within the seed phrase is randomly selected as well as it is randomly selected the feature to be used as a replacement from the Features DB. As stated previously, in order to obtain a language independent algorithm, the system does not have knowledge of part of speech. After the newly generated phrase is evaluated by the system, a mutation rule is created in the form: $A- > B : Value$, where $A$ represents the feature substituted in the seed phrase, $B$ represents the substitution feature (taken from the Feature DB) and *Value* is a weight automatically given to the rule during the evaluation process to determine its usability. Negative value rules are discharged.

### 3.4 Evaluative Criteria

In our approach we attempt to evaluate whether a generated phrase is *correct* through the frequency of appearance of its n-grams in the Web, i.e., the *fitness* as a function of the frequency of appearance. Since matching an entire phrase on the Web might result in very low retrieval, or even non retrieval at all, the sectioning of the given phrase into its respective n-grams is useful. The n-grams frequency of appearance on the Web (using Google search engine) is searched and ranked. The n-grams are evaluated according to the following algorithm: `if` $\alpha < NgramFreq < \theta$, `then` $Ngram$ "weakly accepted"
`elsif` $NgramFreq > \theta$, `then` $Ngram$ "accepted"
`else` $Ngram$ "rejected", `foreach` $Ngram = bigram, trigram, quadrigram$
where, $\alpha$ and $\theta$ are thresholds that vary according to the n-gram type, and $NgramFreq$ is the frequency, or number of hits, returned by the search engine for a given n-gram. The number of n-grams created for a given phrase is automatically determined by the system and it varies according to the length of the phrase. The system evaluates a phrase as "good" if all of its n-grams were labeled "accepted", it is to say, all of the n-grams are above the given threshold. If for a given phrase there is at most *one rejected n-gram* or *one weakly accepted n-gram*, the phrase is evaluated as "usable", otherwise the phrase is "rejected".

## 4 Experiments and Results

The system was setup to perform 1,000 generations for each one, English and Spanish databases. In the case of English, the system generated 2,513 phrases, from which 405 were evaluated as "good", 924 were "usable" and the rest 1,184 were "rejected". In the case of Spanish, there were 1,767 different phrases generated, from which 43 were evaluated as "good", 340 were evaluated as "usable" and the rest 1,384 were "rejected" by the system.

As part of the experiment, the generated phrases were evaluated by a native English speaker and a native Spanish speaker in order to determine their "understandability". By understandability here we refer to the semantic information contained by the phrase, that is to say, how well it expresses information to the

**Table 1.** Human Evaluation: Understandability of the Spanish Phrases

| S. Evaluation | User Evaluation: Semantics | | | User Evaluation: Grammar | | |
|---|---|---|---|---|---|---|
| | 5 | 3 | 1 | 5 | 3 | 1 |
| Good [43] | 18.60% [8] | 60.46% [26] | 20.93%[9] | 30.23% [13] | 32.56% [14] | 37.21% [16] |
| Usable [340] | 12.06% [41] | 49.41% [168] | 38.53% [131] | 13.82% [47] | 39.12% [133] | 47.06% [160] |
| Rejected [1384] | 0.66% [9] | 4.12% [65] | 94.63% [1310] | 0.79% [11] | 4.06% [56] | 95.15% [1317] |

S. = System *[ ] number of phrases

user. We argue that one of the characteristics of human chat is the ability to express semantic information within a given context. This implies that a given phrase does not necessarily have to be grammatically correct in order to be understood and used. A large amount of examples can be found in the chatting rooms on the Web, in English and Spanish. The human evaluation of the generated phrases was performed under the criteria of the following two categories:

a) Grammatical correctness: ranging from 1 to 5, where 1 is incorrect and 5 is completely correct.
b) Semantic correctness: evaluates the ability of the phrase to convey information, asking the question *is the meaning of the phrase understandable?* It ranges from 1 to 5, where 1 is no-understandable, and 5 is completely understandable.

The results of the evaluation are shown in Table 1 and Table 2. Table 1 shows the results for the Spanish phrases. According to the user evaluation of the semantics for the "usable phrases" evaluated by the system, 61.47% (considering the pharses evaluated with 3 to 5 points) of those phrases were semantically understood by the user. The understandability rose considerably, to 79.06%, for the "good phrases". In the case of the Spanish phrases grammar evaluation, the percent of correct grammar for the "good phrases" was around 62.79% while for the "usable phrases" ranked around 52.94%. It is seen here that in many cases (around 18% of the cases) the expresion was understood despite of its grammar.

Table 2 shows the results for the English phrases. For the "good phrases" the understandability ranked around 78.51%, and for the "usable phrases" it ranked 77.05%. On the other hand, the grammar for the "good phrases" is around 96.05% and for the "usable phrases" it is around 98.81%. It is worth noticing that in the case of English the ranking of the grammar was higher than the ranking of the semantics. This is considered to be caused by the appereance of grammatically correct patterns that may not convey semantic information, e.g., [VBP/PRP/VBD/PP/NN], which might be *have you had your lunch?* as well as *have you had your hand?*, being the latter evaluated as grammatically correct (5 points) but semantically incorrect (no-understandable, 1 point) by the user. Therefore in the case of English, in contrast to Spanish, despite of completely correct grammar in many cases lack of meaning arose. On the other hand, the algorithm showed to successfully identify *incorrect phrases*, ranking around 94% in average for the evaluation of grammar and semantics for both English and Spanish.

According to the experiments results, grammatical correctness, however important, does not seem to have a strongly decisive impact on the understandability of

**Table 2.** Human Evaluation: Understandability of the English Phrases

| S. Evaluation | User Evaluation: Semantics | | | User Evaluation: Grammar | | |
|---|---|---|---|---|---|---|
| | 5 | 3 | 1 | 5 | 3 | 1 |
| Good [405] | 34.07% [138] | 44.44% [180] | 21.48% [875] | 39.75% [161] | 56.30% [228] | 3.95% [16] |
| Usable [924] | 19.91% [184] | 57.14% [528] | 22.94% [212] | 13.96% [129] | 84.85% [784] | 1.19% [11] |
| Rejected [1184] | 1.17% [21] | 3.80% [45] | 94.43% [1118] | 3.46% [41] | 3.72% [44] | 92.82% [1099] |

S. = System *[ ]= number of phrases

a trivial dialogue phrase. Therefore, these results reinforce our argument regarding the prevalence of semantics over grammar for trivial dialogue conversation.

## 5 Conclusions and Future Work

In this paper the automatic generation of trivial dialogue phrases was shown through the application of a GA-like transfer approach applied to English and Spanish. The system automatic evaluation of a generated phrase using the WWW as a knowledge database was then judged by a user and a tendency towards the prevalence of understandability over completely correct grammar was observed in the resultant evaluation. As on going work, room is left to explore the application of crossover to the mutation rules, to survey other users opinions in order to identify tendencies, as well as to explore the applicability of the system to Japanese and the validity of the application of the obtained trivial phrases databases to dialogue systems.

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